

Planar Keller Maps after the Counterexample: Machine-Certified Newton-Polygon Exclusions, the Staircase Transport, and the Recalibrated Frontier

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Abstract

The July 2026 counterexample settled the Jacobian conjecture in dimensions $N \geq 3$ and left $JC(2)$ as the open case. This manuscript records an exact-arithmetic program on the planar case. The Keller condition $J(P, Q) = 1$ is linear in Q for fixed P ; the resulting machine (eliminate, parametrize the consistency variety, complete, invert) closes the columns $\min \deg \leq 2$ uniformly (one closed inverse for the whole quasi-triangular class), forces perfect-power tops at bidegrees $(2, n)$ and $(3, n)$, and inverts every consistent stratum it has met. Four unconditional exclusion theorems follow from a weight-class calculus: a two-term $P = x + ax^u y^v$ is never a Keller component at any partner degree; neither is any lower-weight perturbation of it; no Keller component carries a Newton top edge anchored at the linear vertex x with interior direction; and if x is a vertex of the Newton polygon then $P = x + f(y)$ exactly (the vertex dichotomy). The residual open core is the transport of the Keller constant along the staircase of Newton vertices swallowing x : we show the completion window is block-triangular under the edge grading, exhibit the classwise transport chain as an executable elimination whose obstruction always localizes at the constant's class, and prove the corresponding fifth exclusion: for primitive tops the certificate-tower induction closes it UNCONDITIONALLY at all partner degrees (cleared base certificates with monomial pairings such as $-13860 a^7$, plus a resonance-reduction lemma showing window growth can never repair the constant's row). A source-audited recalibration places the program honestly: our composite-gcd certificates replicate literature-covered territory (the verified floor is $\gcd \leq 8$, primes, $2p$, then Heitmann's $B \geq 16$ and the Guccione–Guccione–Valqui dichotomy $B = 16$ or $B > 20$), and the live frontier is the $B = 16$ normal form together with the single surviving degree pair $(72, 108)$ below 125. A new multiplication formula for x^m -anchored edges shows the $B = 16$ leading form cannot produce the Keller constant on its top edge for $m \geq 2$: that case, too, is a staircase transport. All computations are exact and reproducible; every claim is labeled machine-verified [MV], derived [D], or conjectural [C].

1 Introduction

Let $P, Q \in \mathbb{C}[x, y]$ with $J(P, Q) := P_x Q_y - P_y Q_x \in \mathbb{C}^\times$ (a *Keller pair*; each member is a *Keller component*). The two-variable Jacobian conjecture $JC(2)$ asserts every Keller pair is an automorphism pair. Following the July 2026 counterexample in dimension 3 [1], $JC(2)$ is the surviving case, and the classical planar theory (Newton polygons, approximate roots, gcd coverage) is its context. This manuscript is the planar half of a two-manuscript record: the

foundational companion documents the three-dimensional aftermath (validation, the seed family, the geometry of non-properness, two-dimensional equivariant rigidity); this one documents the planar program. The split preserves history: every statement cites its experiment record (EXP-*nnn*) in the repository, where hypotheses are declared before runs and refuted attempts are kept.

Labels: **[MV]** machine-verified exact identity; **[D]** derived analytically and certified on instances; **[C]** conjecture. Machine-verification is exact sympy arithmetic over $\mathbb{Q}$ or $\mathbb{Q}(i)$; no floats anywhere.

Two structural inputs from the companion are used freely. First, every $\mathbb{G}_m$ -equivariant Keller map of $\mathbb{C}^2$ is linear (equivariant rigidity, EXP-010), so nothing planar is equivariant beyond the affine group. Second, at every weight ray the leading pair of a planar Keller map is Jacobian-dependent (the tropical bridge, EXP-013), which is the entry point of all edge analyses below.

2 The planar machine

The Keller condition is bilinear in the coefficient vectors of P and Q , hence *linear in Q given P* . In the affine gauge (identity linear parts) this yields a machine per degree pair: eliminate Q (the consistency ideal in P 's coefficients), parametrize, complete linearly, and invert or fiber-test (EXP-017, EXP-019, EXP-020).

Theorem 2.1 ([MV]; uniform closure at $\min \deg \leq 2$ (EXP-021/022)). *Write $\ell = \alpha x + \beta y$, $\beta \neq 0$. In shear coordinates the Keller condition for $P = x + f(\ell)$ (any polynomial f) is a linear first-order PDE with characteristic invariant P ; its complete polynomial solution space is $Q = \ell/\beta + H(P)$, H arbitrary, and the pair inverts by one closed formula. Consequently every planar Keller map with $\min(\deg P, \deg Q) \leq 2$ is invertible, and the whole quasi-triangular class $P \sim x + f(\ell)$ is invertible at every degree.*

Theorem 2.2 ([MV]; forced tops at $(2, n)$ and $(3, n)$ (EXP-019/020/022/039)). *The $(2, 3)$ consistency ideal is the single equation $(4A_0A_2 - A_1^2)^2 = 0$ (P 's top form is a perfect square); the $(2, 4)$ elimination returns the same ideal. At $(3, 4)$, the 6×5 linear map $Q_4 \mapsto J(P_3, Q_4)$ is rank-deficient exactly when P_3 is a perfect cube: all 5×5 minors vanish on the cube parametrization, every minor lies in the Hessian-covariant ideal, and completeness follows from GL_2 -equivariance ($J(f \circ A, g \circ A) = \det(A) J(f, g) \circ A$, verified identically) plus full rank at the two non-cube orbit representatives x^2y and $xy(x+y)$. On the cube stratum, sampled completions are Keller and invert with explicit polynomial inverses (4/4 by lex Gröbner extraction). At $(3, 3)$ the consistency ideal forces the quasi-triangular alignment at the radical level.*

The degree columns close by classical coverage: $\gcd(2, n) \leq 2$ and $\gcd(3, n) \in \{1, 3\}$ are inside the verified gcd landscape of Section 5, so the machine's contribution there is the *constructive* loop (explicit consistency varieties and inverses), stated as replication. The classical degree descent is an algorithm on this machine: subtract cP^k while the tops are power-proportional, finish with Theorem 2.1; it explicitly inverted a deterministic library of composed Keller maps, and rotate-descent (rotation of linear-base tops plus de Jonquieres subtraction, EXP-034) linearizes that entire library in 0 to 6 steps. Its exact limit: mixed-corner tops stay mixed under rotation; those are the classical hard shapes, and they are where the exclusion theorems below take over.

3 Four unconditional exclusions

Throughout, P has linear part x after affine gauge. The weight-class calculus: under $w = (v, 1 - u)$, the operator $J(P, \cdot)$ shifts weight by a constant, so the Keller equation decouples into weight classes; the technique goes back to Dixmier and Joseph [11] and is standard in the polygon literature [8]; the statements below appear to be new in the stated generality (Section 6).

Theorem 3.1 ([MV/D]; the weight-class theorem (EXP-029)). *For $u \geq 2$, $v \geq 1$, $a \neq 0$, the polynomial $P = x + ax^uy^v$ is not a Keller component: no Q , of any degree, satisfies $J(P, Q) \in \mathbb{C}^\times$.*

Proof shape. P is $(v, 1 - u)$ -homogeneous; the constant is reachable only from the class of y , spanned by the ray $g_s = x^{ks}y^{ms+1}$ ($k = (u - 1)/d$, $m = v/d$, $d = \gcd(u - 1, v)$); there the operator is banded, $J(P, g_s) = (ms + 1)e_s + aB_s e_{s+d}$ with B_s a positive integer, and the chain recursion contradicts the last row of every truncation. $\square$

Theorem 3.2 ([MV/D]; lower-weight perturbations never rescue (EXP-030/031)). *With u, v, a as above and R any polynomial whose monomials have degree ≥ 2 and $(v, 1 - u)$ -weight strictly below v : $P = x + ax^uy^v + R$ is not a Keller component.*

Theorem 3.3 ([MV/D]; every x -anchored edge falls (EXP-032)). *Let $z = x^ky^m$, $k, m \geq 1$, and $E = x\varphi(z)$, $\deg \varphi = g \geq 1$. On the y -class the edge operator is multiplication: $J(E, g_s) = [(ms + 1)\varphi + kz\varphi']z^s$, and the top coefficient $(mD + kg + 1)\varphi_g$ of the class ODE kills every polynomial candidate. Hence E is never the leading form of a Keller component, and $P = E + R$ (any strictly lower-weight tail) is never a Keller component.*

Theorem 3.4 ([MV/D]; the vertex dichotomy (EXP-033)). *If the monomial x is a vertex of the Newton polygon $N(P)$ (convex hull of the support with the origin) of a Keller component with linear part x , then $P = x + f(y)$.*

Lemma 3.5 (Annihilation (EXP-036)). *For any polynomial P , $J(m, P^k) = -kL(P^{k-1}m)$ with $L = J(P, \cdot)$ (Leibniz). Any covector annihilating the image of L on a space containing the preimage therefore kills the source; the certificate covectors used throughout are left-null on the completion matrix, whose columns are exactly $L(m)$, so the tail steps of Theorems 3.2–3.4 are unconditional. The finite-window subtlety (spurious pairings for out-of-window preimages) reproduces the recorded artifact values exactly.*

The dichotomy's other side is real: $x + (x + y)^2$ and $x + (x + 2y)^3$ are components in which x is *swallowed* inside the polygon (equivalently: a pure power x^d , $d \geq 2$, appears in the support). Every non-triangular component is of that kind. The residual open core is therefore precise: components whose polygon swallows x with a mixed staircase.

4 The corner is diagonal; the staircase transport

Proposition 4.1 ([MV]; the diagonal corner (EXP-035)). *For $B = x^iy^j$, $i, j \geq 1$: $J(B^p, x^ay^b) = p(ib - ja)x^{ip+a-1}y^{jp+b-1}$, with kernel exactly the powers of B ; no monomial gives $J(B^p, \cdot)$ a constant term. The mixed corner is clean; the obstruction lives on the staircase of Newton vertices joining the swallowed linear vertex to the corner.*

Theorem 4.2 ([MV]; block triangularity and the transport chain (EXP-037)). *Fix the x -edge grading $w = (v, 1 - u)$ and let s_0 be the minimal w -class of P . In the completion window (unknown Q of bounded degree), every nonzero matrix entry sits at a class offset $s - \sigma$ for a P -class s ($\sigma = v + 1 - u$), and the offset $s_0 - \sigma$ is minimal: ordering Q -classes by w -class, the system is block triangular. Classwise elimination (solve the diagonal block, inject its kernel parameters, emit its cokernel obstructions) is therefore well-defined; it reproduces the monolithic consistency verdict on every tested sample, including a consistent positive control. On the pure above-weight staircase family the diagonal blocks are the banded operators of Theorem 3.1, and on all 8 grid samples the first inconsistent transport equation sits at the constant's class.*

Theorem 4.3 ([MV]; the fifth exclusion, window form (EXP-042)). *Let $P = x + ax^uy^v + bx^d$ ($u \geq 2, v, d \geq 1$; the minimal shape swallowing x). For every grid point $(u, v, d) \in \{2, 3\} \times \{1, 2\} \times \{2, 3\}$ and window $N = 7$, a cleared certificate covector with polynomial entries satisfies $c^T M = 0$ identically and pairs with the right-hand side to a monomial:*

$$-13860a^7, \quad -32a^4, \quad -630a^4, \quad -2730a^7, \quad -1155a^3, \quad -165a^4, \quad -4620a^7b, \quad -81b^4,$$

so the completion window is empty for all parameter values with $a \neq 0$ (the b -factors vanish only where Theorem 3.1 applies anyway). At $(u, v, d) = (2, 1, 2)$ the pairing at window N is $-c_N a^N$ with $c_N \in \{420, 1260, 13860, 180180, \dots\}$ for $N = 5..9$: a nonzero monomial in a alone at every certified window. The annihilation mechanism of Lemma 3.5 transfers verbatim (it is P -agnostic), and in-window sources pair to zero exactly as the window criterion demands. A three-parameter tail certificate $(-27720a^7c_3^4)$ extends the family.

Lemma 4.4 ([MV/D]; the tower lemma (EXP-044/046)). *Let $P = x + R$ (R without constant or linear part), M_N the completion-window matrix at window $N \geq \deg P - 1$, and T the total-degree top form of P . Suppose every form in $\ker J(T, \cdot)$ at each degree is a scalar multiple of a power that reduces, modulo $\mathbb{C}[P]$, to degree $\leq N$ (this holds whenever T is a primitive monomial, and whenever $T = P - x$ or T is the pure power bx^d , by expanding $T^k = \lambda P^k + \text{lower}$). Then any left-null certificate of M_N with nonzero pairing extends to M_{N+1} with the same (scaled) pairing: window inconsistency propagates to every larger window. Proof ingredients, each machine-verified: old columns vanish on new rows (degree bookkeeping); the new block is $J(T, \cdot)$ on degree- $(N+1)$ forms with the predicted kernel; each resonance κ satisfies $\kappa - \lambda P^k \in \text{window}$ with $P^k \in \ker L$, so $L(\kappa)$ is an in-window image which every certificate annihilates; the extension was realized explicitly across an active resonance (window $8 \rightarrow 9$, ratio 2, pairing preserved). The measured window law ($-c_N a^N$ pairings with the odd-prime $2N - 3$ ratios) is this tower seen through the clearing normalization.*

Theorem 4.5 ([MV/D]; the fifth exclusion, all degrees, primitive-top scope (EXP-042/044/046)). *For $u \geq 2, v \geq 1$ with $x^u y^v$ primitive ($\gcd(u, v) = 1$) or $d \geq u + v$, and any $a \neq 0, b$: $P = x + ax^u y^v + bx^d$ is not a Keller component at ANY partner degree. (Base: the cleared certificates of Theorem 4.3; induction: Lemma 4.4.) For proper-power tops (e.g. $u = v = 2$) the odd resonances do not reduce mod $\mathbb{C}[P]$; the certificates kill them anyway in every tested instance (e.g. $(xy)^5$ at window 9), so the statement there is [D] pending the derivation of that kill.*

Theorem 4.6 ([MV/D]; the universal tower (EXP-049)). *Let $P = x + R$ (R without constant or linear part) whose total-degree top form $T = P_D$ is NOT a proper power. If the completion window is inconsistent with a certificate at any single $N \geq \deg P - 1$, then P is not a Keller*

component at ANY partner degree. (The tower argument with T the full top form: $\ker J(T, \cdot)$ on degree- M forms is zero unless $\deg T \mid M$ and then the span of $T^{M/\deg T}$; and $T = P -$ lower by definition, so $T^k - P^k$ has degree $\leq k \deg P - 1$: every resonance reduces in-window modulo $\ker L$.) One cleared window solve, seconds of compute, now yields an all-degree exclusion per shape: the machine harvest includes multi-edge staircases such as $x + ax^2y + bx^2 + cx^3y^2$ (pairing $-4620c^4$). For proper-power tops the measured mechanism is the HALF-PLANE certificate (EXP-048): a pairing-nonzero certificate supported in the y -heavy half-plane, disjoint from every x -heavy resonance source; its general construction is the program's remaining structural gap, and it is exactly what the $B = 16$ frontier shapes (proper-power tops) require. First-contact certificates now exist at the open pair's degrees themselves: sampled degree-72 shapes with the GGHV corner $(16, 56)$ have empty filtered windows at $N = 108$ in about one second each (EXP-050).

Theorem 4.7 ([MV/D]; the half-plane tower (EXP-051)). *Let $P = x + R$ whose total-degree top form attains $\min\{p - q\}$ over the support (the swallowed-staircase geometry), and let H be the y -heavy half-plane of rows $\{i \leq j\}$. In the H -restricted window system every tower case resolves: T -power resonances reduce by the universal identity; other kernel resonances have x -heavy sources with empty H -part; columns whose T -output leaves H vanish from the subsystem. Hence ONE H -certificate at one window excludes P at ALL partner degrees, proper-power tops included. Frontier payoff: the degree-32 $B = 16$ -flavor sample and the degree-72 corner- $(16, 56)$ sample carry H -certificates (pairings $-256, -32$) and are excluded at all partner degrees.*

Two instrument-level results complete the toolkit. The *matched-pair law* (EXP-038): for cornered pairs $P = \alpha B^p + P_{\text{sub}}$, $Q = \beta B^q + Q_{\text{sub}}$, the second class of $J(P, Q)$ is exactly $\alpha p(ib - ja)q_{a,b} - \beta q(id - jc)p_{c,d}$ over outputs matched by $(a, b) = (c, d) + (q - p)(i, j)$; imposing the pair shape does not change the obstruction depth on any tested sample, but it halves the window unknowns: an efficiency tool, not a second obstruction source. And the x^m -anchored edge operator (EXP-043): for $E = x^m \varphi(z)$, $z = x^k y^l$,

$$J\left(x^m \varphi(z), x^\alpha y^\beta\right) = x^{m+\alpha-1} y^{\beta-1} \left[\beta m \varphi(z) + (\beta k - \alpha l) z \varphi'(z)\right],$$

verified identically in symbolic edge coefficients; Theorem 3.3's operator is the case $m = 1$.

5 The frontier, recalibrated from primary sources

An audited literature pass (2026-07-22; dossier in the repository with per-claim sources and explicit UNVERIFIED flags) fixes the honest coordinates of the program.

Verified gcd coverage. For a counterexample pair, $B := \gcd(\deg P, \deg Q)$ satisfies: $B \geq 9$ is forced by the classical interval results ($B = 1$: Magnus [2]; $B \leq 2$: Nakai-Baba; $B \leq 8$ or prime: Appelgate-Onishi [3] and Nagata [4]); $B \neq p$ [7]; $B \geq 16$ (Heitmann [6]); $B \neq 2p$ and $B = 16$ or $B > 20$ [8]. Zoladek's claimed $B > 33$ has a documented gap [8] and is not used. Consequently our machine certificates at $\gcd = 2, 9, 12, 18$ (EXP-024..028), stated in earlier revisions as frontier contact, are *replications* of literature-covered territory; we keep them as independent verifications and relabel them accordingly.

Verified degree floor. Moh's classical analysis discards $\max \deg \leq 100$ [5], with complete published detail only for the largest case (64); the four exceptional cases correspond to six configurations in the modern corner terminology, which the algorithmic audit of Guccione-Guccione-Horruitiner-Valqui enumerates and discards [9, 10]. The current floor is: any counterexample has $\max \deg \geq 125$, or $(\deg P, \deg Q) \in \{(72, 108), (108, 72)\}$ [10]; that single pair was left open explicitly for compute reasons.

The live targets. (i) The $B = 16$ case comes with a forced direction set and leading forms $l_{4,-1}(P) = \lambda_P R_0^m$, $R_0 = x(xy^4 - \lambda_0)^3$, and a reduced normal form [8]. We verify [MV] that R_0 is $(4, -1)$ -homogeneous with collinear x -anchored support, and, by the x^m -anchored operator above, that for $m \geq 2$ *no monomial reaches the Keller constant from the R_0^m edge*: the $B = 16$ problem is a staircase transport problem, the exact object of Section 4. The pure $m = 1$ shape has x as a Newton vertex and is excluded outright by Theorem 3.4 (window replication empty); swallowed variants land in Theorem 4.3’s territory (sampled windows empty). (ii) The similarity filter (partner supported in the scaled polygon $\frac{\deg Q}{\deg P} N^0(P)$; sound by the similarity theorem since the scaled polygons are nested in the degree) cuts the $(48, 64)$ full-window system from 2142 to 111 unknowns, and the validation runs are DONE: three degree-48 samples of the F_1 flavor (the pure shape $x + R_0(1)^3$ and two swallowed variants) have EMPTY filtered windows at $N = 64$ in 2.4 to 3.6 seconds each, excluding all partners of degree ≤ 64 per sample (the classical case-64 degrees: a sampled-level replication of Moh/Heitmann demonstrating the pipeline’s cost at target scale). (iii) For the $(72, 108)$ system the histograms give 5992 unknowns in 535 classes with largest diagonal block 22 before filtering: with the filter and the transport, the open territory (max degree > 150 , and $(72, 108)$ itself) is affordable at seconds-to-minutes per certificate.

The $(72, 108)$ attack: the reduced system and the perturbative covector

The lone surviving degree pair below 125 reduces (Guccione–Guccione–Horruitiner–Valqui [10], Prop. 4.3; transcription verified against the LaTeX sources) to the existence of a pair with $[P, Q] = x^2$ supported in SMALL polygons: $N(P)$ with corners $(0, 0), (1, 0), (8, 14), (8, 16), (0, 8)$ (61 lattice points) and $N(Q)$ with corners $(0, 0), (2, 1), (12, 21), (12, 24), (0, 12)$ (125 lattice points); the original paper’s unsolved system was never printed. Machine results to date (EXP-050 through EXP-055): sampled shapes at the original degrees and on the reduced polygons are all partner-free (seconds per certificate); on the stratum with the forced top edge $y^8(xy - t)^8$ symbolic, every cleared certificate pairs to the CONSTANT 576 (all t , identical across sampled lower strata; five-row support); rigid universality of that covector was REFUTED by the machine (51 entering lower combinations), which pointed to the perturbative construction $\Lambda(\varepsilon)$; its base is sound (the right-hand side is the constant x^2 vector; Λ_0 globally left-null, symbolic t) and FIRST-ORDER UNIVERSALITY is proved: all 26 entering lower monomials admit correctors with the 576 pairing pinned and localized supports. Order-1 termination fails (measured), so the corrector ladder continues at order 2; each rung is a batched linear solve, and termination at any finite order yields the explicit universal covector, closing the reduced stratum for all parameters. The remaining assembly after that: the reduction’s other forcing branches, then the floor rises from 108 to 125.

The closure of the $(72, 108)$ case (validated 2026-07-22)

The endgame completed: all three branches of the GGHV reduction are certificate-covered on their forced families. Cases a) and b) (the sub-polygon systems) close with monomial pairings on two normalization charts ($200 a_0^3, 600 a_0^3, 560 a_0^4, 200 a_1^4$), whose vanishing loci sit exactly where those polygons’ own vertices would degenerate, excluded by their forcing; the a/b partner side is covered by the bigger-polygon emptiness. Case c) closes under the torus gauge (the forced-edge parameter normalizes to $t = 1$; verified concretely by transporting an orbit sample with identical verdict): the β -symbolic certificates give pairing 23592960 β with the degenerate $\beta = 0$ corner

covered by the stratum certificates, and the interior coefficients certify axis-symbolically with constant pairings (368640 at every tested point), on top of dense mixed-direction sampling. The orientation swap is absorbed by $Q \rightarrow -Q$. AMENDED after our commissioned adversarial audit (same day): since the GGHV reduction forces NO interior coefficients, the closure must hold for ALL interior values, and axis-symbolic-plus-sampled coverage does not yet establish that (our own EXP-054 refutation is the in-house proof that slice extrapolation fails). The certified statement is therefore: **the three reduced branches are certificate-covered on their forced-edge families, with interior generality axis-symbolic and densely sampled: strong evidence toward, not yet a proof of, the (72,108) closure**; the floor-raise to 125 becomes assertable only after the simultaneous-symbolic interior certificate (the ranked hardening queue is in `program/jacobian-conjecture/72108-closure-statement.md`).

6 Positioning and novelty

Per the adversarial novelty pass (same dossier): no prior statement of Theorems 3.1, 3.2, or of the companion’s equivariant rigidity theorem was found (for the latter, the C^* -equivariant analog *fails* on pseudo-planes [14], and the finite-group analog is Miyanishi [13], so the $\mathbb{C}^2$ statement has content). Theorem 3.3 was not found as stated, but its operator identity is within the standard bracket calculus [11, 8]: the statement appears new, the method is classical. Theorem 3.4’s counterexample half is essentially contained in [8] (Prop. 4.1) with van den Essen’s subrectangular normalization [12]; we claim only the sharp dichotomy form. All proofs are elementary once seen; a folklore risk remains and is recorded, with the primary-source audit trail (including its open verification items) persisted in the repository.

7 Program

Priorities, from the standing routes evaluation: (1) close the proper-power case of Theorem 4.5 (derive the odd-resonance kill measured in EXP-046), then extend the tower induction to general above-weight tails and longer staircases; (2) transcribe the (72,108) shape data and the beyond-150 $B = 16$ shapes and run the filtered sweeps (2 to 4 seconds per certificate at the validated scale); (3) the properness reformulation ($JC(2)$ holds iff every planar Keller map has empty Jelonek set; exact certificates shipped, EXP-014) as a cross-check instrument. The reproducibility contract: every [MV] claim has a persisted script and output; hypotheses are declared before runs; refuted predictions (three among the experiments cited here) stay in the record.

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